

KISHORE KUMAR RAMKUMAR

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EDUCATION

Indiana University Bloomington, Indiana
Master of Science, Data Science ; GPA - 3.5

Aug 2024 - May 2026
Indiana

Sri Ramachandra Institute of Higher Education and Research, Chennai
Bachelor of Technology, Computer Science Specializing in AI & ML

Jul 2021 - Jun 2024
Chennai

TECHNICAL SKILLS

- **Programming Languages & Development:** C/C++, Python, TypeScript, Java, SQL, JavaScript
- **Core CS:** Data Structures, Algorithms, Object-Oriented Design, Multithreading, Linux
- **Distributed Systems:** Microservices, Job Scheduling, Fault Tolerance, Parallel & Distributed Computing, ETL Pipelines
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, DynamoDB, AWS S3
- **Cloud & DevOps:** AWS (EC2, Lambda, EKS, SQS, CloudWatch), Docker, Kubernetes, Terraform, CI/CD, Git
- **Additional Experience (ML Systems):** RAG/FAISS, PyTorch, SageMaker, Bedrock, Model Serving, TensorFlow

EXPERIENCE

Indiana University O'Neill School of Public & Environmental Affairs
Research Assistant (Data Engineering)

May 2025 - Aug 2025
Bloomington, IN

- Built and operated a **distributed data processing pipeline** to process **530+ GB** of CMIP6 precipitation data, regridded to a $1^\circ \times 1^\circ$ global grid using xarray and xESMF, and published versioned NetCDF outputs for cross-model analysis.
- Reduced preprocessing time by **32%** on Quartz **HPC** by distributing workloads and optimizing I/O operations to improve **Slate** storage throughput.
- Generated 3, 6, and 12 month rolling aggregates and CONUS maps, adding **data validation** and drift checks to support reliable **time series analysis** for drought indicators.

MRAR Intellect Solutions
Machine Learning Associate Engineer Intern

Jan 2024 - Aug 2024
Mississauga, Canada

- Built and deployed **backend microservices** for a retrieval-augmented generation (**RAG**) **chatbot** using FastAPI, FAISS-GPU, and PyTorch on **AWS**, enabling faster query responses and improving answer relevance.
- Deployed inference services on **AWS EKS** behind an **ALB** with **autoscaling** and **CloudWatch** monitoring, keeping **p95 latency under 150 ms**.
- Implemented **CI/CD pipelines** with **Docker** and **ECR** and added CloudWatch logs and dashboards, increasing GPU evaluation throughput by **18%**.
- Designed **fault-tolerant inference** workflows with safe retries and health checks to handle pod restarts and transient failures in a distributed environment, reducing service interruptions and improving overall system reliability.

IVIEWSENSE Private Limited
Data Analyst Intern

Jan 2022 - Dec 2022
Chennai, India

- Collaborated with **business teams** to define **key performance indicators** and questions, and built interactive **dashboards** in Microsoft **Power BI** using **SQL** and Data Analysis (**EDA**), enhancing data driven decision making.
- Streamlined reporting with scheduled refresh, data validation, and monitoring alerts, reducing refresh time from 42 minutes to 33 minutes and keeping updates **98% on time**.

PROJECTS

TruthLens (AWS AI Agent Hackathon) | <https://github.com/kishuxz/aws-hackathon-truthlens>

- **Led** a team to design and built a **multimodal** reliability agent on **AWS Bedrock** (Nova Micro) that outputs an auditable 0–100 score with labels, citations, independence checks, and contradiction checks.
- Built a **React** web app + **serverless API (Lambda Function URL, S3)** with CloudWatch metrics/logging, **IAM** least privilege, and **Pydantic validation**; met latency/SLO expectations.
- Delivered an end to end pipeline with **SageMaker OCR** for quote cards and planned **Bedrock AgentCore** orchestration and **policy driven scoring**.

Cloud Job Scheduler (AWS) | <https://github.com/kishuxz/Job-Scheduler>

- Built a cloud native **distributed job scheduling** service using **AWS Lambda, SQS, DynamoDB, and ECS**.
- Implemented **idempotent job** submission to safely handle client retries and prevent duplicate execution in a distributed system.
- Designed **fault tolerant job execution** with asynchronous workers and explicit state transitions (PENDING, RUNNING, SUCCEEDED, FAILED).
- Owned deployment, monitoring, and troubleshooting using CloudWatch logs and metrics to ensure service reliability.

NetworkMap (Open-Source Network Visualization Platform) | https://github.com/kishuxz/info_viz

- Built a **network visualization platform** to model and explore relationships between entities, using a **Node.js** backend and **MongoDB** for persistent graph data storage.
- Implemented an **automated data ingestion pipeline** to parse, validate, and transform CSV inputs into structured node/edge datasets.
- Designed **API driven** data loading and querying to support efficient exploration of **large graph datasets at scale**.

CERTIFICATIONS

- AWS Certified : Machine Learning Engineer - Associate : November 2025 - November 2028
- AWS Certified : Developer - Associate : January 2026 - January 2029